

班号 _____ 姓名 _____ 学号 _____

2016-2017 第一学期参考答案

每题 5 分). (1) 由 $w = 2(3+i)$ 得 $z = \frac{w}{2} - i$, 将其代入 $|z-i| < 1$ 有 $|w-4i| < 2$. (5分)

由 $f(z) = xy^2 - x^2y$ 得 $\begin{cases} u(x,y) = xy^2 \\ v(x,y) = -x^2y \end{cases}$ 从而 $u'_x = y^2, u'_y = 2xy, v'_x = -2xy, v'_y = -x^2$. (3分)

由柯西-黎曼条件 $\begin{cases} u'_x = -v'_y \\ u'_y = v'_x \end{cases}$ 得 $\begin{cases} y^2 = -x^2 \\ 2xy = -2xy \end{cases}$ 即 $x^2 + y^2 = 0$, 因此 $f(z)$ 仅在 $z=0$ 处可导, 由柯西-黎曼条件的定义知 $f(z)$ 无解析点. (4分)

对于 $u(x,y) - v(x,y) = x^2y^2 - 2xy$ 关于 x, y 分别求偏导得 $\begin{cases} u'_x - v'_x = 2x - 2y \\ u'_y - v'_y = -2y - 2x \end{cases}$ (2分)

由 $\begin{cases} u'_x = v'_y \\ -u'_y = v'_x \end{cases}$ 代入并解方程得 $\begin{cases} u'_x = 2x \\ u'_y = -2y \end{cases}$ 从而 $f'(z) = u'_x - u'_y i = 2z$ (4分)

故 $f(z) = z^2 + C(1+i)$, 其中 C 为实常数. (6分)

(6分) 第一小题 12 分, 其它儿小, 题每题 6 分).

1) 直线段的参数方程 $z(t) = (2+i)t, t \in [0,1]$, 从而 $\operatorname{Re}(z) = 2t, dz = (2+i)dt$. (2分)

$\int_C z dz = \int_0^1 2t(2+i)dt = 2+i$. (6分)

2) $z=0$ 到 $z=i$ 的直线段 C_1 的参数方程为 $z(t) = ti, t \in [0,1]$, $z=i$ 到 $z=2+i$ 的直线段 C_2 的参数方程为 $z(t) = 1+i, t \in [0,2]$. 于是 $\int_C \operatorname{Re}(z) dz = \int_0^1 0 \cdot i dt + \int_0^2 1 dt = 2$. (6分)

3) $\int_{|z|=1} \frac{dz}{(z^2+1)^2} = \frac{2\pi i}{1!} \left[\frac{1}{(z^2+1)^2} \right]' \Big|_{z=-1} = 2\pi i \cdot \frac{2}{9} = \frac{4\pi i}{9}$. (或者 $= 2\pi i \operatorname{Res}\left[\frac{1}{(z^2+1)^2}, -1\right] = \dots$) (6分)

4) $\int_{|z|=2} \frac{ze^z}{z^2-1} dz = 2\pi i (\operatorname{Res}\left[\frac{ze^z}{z^2-1}, 1\right] + \operatorname{Res}\left[\frac{ze^z}{z^2-1}, -1\right]) = 2\pi i \left(\frac{e}{2} - \frac{e}{2} \right) = 0$. (6分)

5) $\int_{|z|=2} \frac{z^3}{(z^2+1)^2} dz = 2\pi i (\operatorname{Res}\left[\frac{z^3}{(z^2+1)^2}, i\right] + \operatorname{Res}\left[\frac{z^3}{(z^2+1)^2}, -i\right]) = 2\pi i \left(\left(\frac{z^3}{(z^2+1)^2} \right)' \Big|_{z=i} + \left(\frac{z^3}{(z^2+1)^2} \right)' \Big|_{z=-i} \right) = 2\pi i \left(\frac{1}{2} + \frac{1}{2} \right) = 2\pi i$. (6分)

6) $\int_{|z|=4} \frac{dz}{(z-3)(z^2-1)} = \frac{2\pi i}{1!} \left[\frac{1}{(z-3)(z^2-1)} \right]' \Big|_{z=3} = \frac{\pi i}{12}$. 而 $\int_{|z|=4} \frac{dz}{(z-3)(z^2-1)} = \int_{\gamma=4} \frac{1}{(\gamma^2-3)(\gamma^2-1)} \frac{1}{\gamma^2} d\gamma = \int_{\gamma=4} \frac{\gamma^4 d\gamma}{(\gamma^2-3)(\gamma^2-1)\gamma^2}$. (4分)

7) $\int_{|z|=1} \frac{dz}{(z-2)(z^2-1)} = -\frac{\pi i}{10}$. (2分)

$$\int_{-\infty}^{+\infty} \frac{x}{5-4\cos x} dx = \int_0^{2\pi} \frac{dx}{5-2(e^{ix}+e^{-ix})} = \int_{|z|=1} \frac{1}{5-2(z+z^{-1})} \frac{dz}{iz} = i \int_{|z|=1} \frac{dz}{(2z+1)(z-2)} = \frac{i}{2} \int_{|z|=1} \frac{\frac{1}{z-2}}{z-\frac{1}{2}} dz$$

$$= \frac{i}{2} \cdot 2\pi i \cdot (-\frac{2}{3}) = \frac{2}{3}\pi \quad (6分)$$

$$(7) \int_0^{+\infty} \frac{x^2}{(x^2+1)^2} dx = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x^2}{(x^2+1)^2} dx = \frac{\pi}{2} - \frac{1}{2} \int_{-\infty}^{+\infty} \frac{dx}{(x^2+1)^2} = \frac{\pi}{2} - \pi i \operatorname{Res}\left[\frac{1}{(z^2+1)^2}, i\right] = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \quad (6分)$$

$$(8) \int_{|z|=1} \frac{1}{z \sin z} dz = 2\pi i \operatorname{Res}\left[\frac{1}{z \sin z}, 0\right] \quad (2分)$$

由于 $\sin z$ 关于 z 的偶函数, 所以 $\frac{1}{z \sin z}$ 在 $0 < |z| < 1$ 的罗伦展开式中仅出现偶次幂项, 从而 $C_{-1} = 0$ (4分)

$$= 2\pi i C_{-1} = 0 \quad (6分)$$

$$(9) \int_{|z|=2} \frac{z^n}{1+z^n} dz = \int_{|z|=\frac{1}{2}} \frac{dz}{1+z^{n+1}} = 2\pi i C_{-1} \quad (2分)$$

$\frac{1}{1+z^{n+1}}$ 在 $0 < |z| < 1$ 的罗伦展开式为 $\sum_{k=0}^{+\infty} (-1)^k z^{k(n+1)}$

$$= \sum_{k=0}^{+\infty} (-1)^k z^{nk+n+1} \quad nk+n+1 = -1 \Rightarrow nk = -n-2 \Rightarrow n \mid (n+2) \Rightarrow n \mid 2$$

可见当 $n \neq 1$ 时 $C_{-1} = 0$, 当 $n=1$ 时 $C_{-1} = (-1)^2 = 1$ (4分)

$$= \begin{cases} 0, & n \neq 1 \\ 2\pi i, & n=1 \end{cases} \quad (6分)$$

$$(1) f(z) = \frac{1}{z} = \frac{1}{\frac{1}{2} + \frac{1}{2}i} = \frac{1}{\frac{1}{2}(1+i)} = \frac{2}{1+i} = \frac{2(1-i)}{(1+i)(1-i)} = \frac{2(1-i)}{2} = 1-i \quad (2分)$$

$$(2) f(z) = \frac{1}{z^2(1+z)} = \frac{1}{z^2} \sum_{k=0}^{+\infty} (-1)^k z^k = \sum_{k=0}^{+\infty} (-1)^k z^{k-2} = \sum_{k=-2}^{+\infty} (-1)^{k+2} z^k \quad (4分)$$

$$(2) \text{ 当 } |z+1| > 1 \text{ 时 } \frac{1}{z} = -\left(\frac{1}{z}\right)' = -\left(\frac{1}{z+1}\right)' = -\left(\frac{1}{z+1} \cdot \frac{1}{1-\frac{1}{z+1}}\right)' = -\left(\sum_{k=0}^{+\infty} \left(\frac{1}{z+1}\right)^{k+1}\right)'$$

$$= -\sum_{k=0}^{+\infty} (k+1) (z+1)^{-k-2} \quad (4分)$$

$$\text{于是 } f(z) = \sum_{k=0}^{+\infty} (k+1) (z+1)^{-k-2} = \sum_{k=-\infty}^{-3} (k+2) (z+1)^k \quad (6分)$$

(6分) 由 $u(x,y) = v(x,y)$ 得 $\begin{cases} u'_x = 2v v'_x \\ u'_y = 2v v'_y \end{cases}$ 将 C-R 条件 $\begin{cases} u'_x = v'_y \\ -u'_y = v'_x \end{cases}$ 代入得 $\begin{cases} u'_x = -2v u'_y \\ u'_y = 2v u'_x \end{cases} \quad (4分)$

于是 $u'_y = 2v \cdot (-2v u'_y) = -4v^2 u'_y$ 即 $(1+4v^2) u'_y = 0$ 从而 $u'_y(x,y) = 0, u'_x(x,y) = 0$

故 $u(x,y)$ 是个常数, 从而 $f(z)$ 必为常数. (6分)